

BRYAN SAMUEL JAMES

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Available for Fall 2026, Spring 2027, and Summer 2027 Internships/Co-ops

EDUCATION

Master of Science, Computer Science – Northeastern University, Boston, MA

Sep 2025 – Dec 2027

Relevant coursework: Database Management Systems, Essentials of Data Science, Programming Design Paradigm, Foundations of Artificial Intelligence.

Bachelor of Technology, Information Technology – Vellore Institute of Technology, India

Jul 2021 – May 2025

Relevant coursework: Data Structures and Algorithms, Operating Systems, Software Engineering, Artificial Intelligence, Probability and Statistics.

EXPERIENCE

Software Engineering Intern – Infyz Solutions Pvt Ltd (Aug 2023 – Jan 2024)

- Reduced manual data-entry delays by **30%**, measured by workflow completion time, by building Java-based REST API modules automating "Inward" and "Issue" workflows on the ITOMS platform.
- Improved cost-control accuracy by **25%**, measured by reporting error rate, by building SQL-based ABC and Aging Analysis reports using JDBC.
- Increased release reliability, measured by reduced sync failures, by building automated alert services and shipping features via Git/Docker-based Agile sprints.

PROJECTS

AgentFirst – MCP-Based AI Commerce Agent (React/TypeScript)

Jun 2026

- Built an AI agent using MCP as part of a team project, that browses, reasons over what it finds, and completes a full transaction on its own inside ChatGPT, rather than a scripted demo.
- Shipped a full-stack app end to end, designing REST API endpoints (TypeScript/Bun) with a React frontend, Stripe payments, and a PostgreSQL database, with automated tests validating reliability across every layer.

Patent Search Engine – Hybrid Semantic Search with Reranking

Aug 2026

- Built a hybrid search engine over **10,578** patent claims, combining SQL pre-filtering with sentence-transformer embeddings indexed in **FAISS** for semantic vector search, cross-referencing candidates through a cross-encoder reranking stage, served through a FastAPI endpoint.
- Empirically caught a correctness bug in a naive filter-after approach (missed **40 of 41** true matches) that a filter-first design avoided entirely, evidence over assumption in choosing the architecture.

Parity-Agent – Dynamic Pricing Detection Agent (TrueForge/MCP)

Aug 2026

- Built an agent on the TrueForge harness that detects personalized and dynamic pricing by fetching the same product from isolated, different-country sessions and diffing the normalized prices, using Bright Data (wrapped as a custom MCP server) for multi-session fetching.
- Verified two real cases: a correct negative on Amazon (no IP-based pricing) and a positive on Steam (~**99%** regional price spread, reproducible); built self-healing selector repair that detects broken scrapers, has a model relocate the price, and validates the fix before persisting it.

Contract-Guard – Schema-Contract-Enforced Data Ingestion

Aug 2026

- Built dual ingestion pipelines (naive vs. contract-validated) in Python with Airflow and dbt, quarantining schema drift from multiple source systems before it reaches downstream reporting.
- Proved effectiveness with a controlled 200-trial experiment: **100%** catch rate on breaking schema drift, **0%** false positives, and **+15.5ms** median overhead.

PUBLICATIONS

Thyroid Disease Prediction Using ML Models (Book Chapter, Jan 2025).

SKILLS

Languages: Python, Java, SQL, JavaScript, TypeScript.

AI Engineering: Generative AI, Agentic AI, LLM Orchestration, LangChain, LangGraph, Prompt Engineering, RAG, Tool-Calling (MCP), Evals, Sentence-Transformer Embeddings, FAISS, Semantic Search, Reranking, scikit-learn.

Data & Backend: REST API Design, FastAPI, React, Node.js/Bun, PostgreSQL, SQLite, AWS.

AI Tools: Claude Code, Qodo (AI Code Review), Cursor.

DevOps & Practices: Git, Docker, Kubernetes, CI/CD (GitHub Actions), Airflow, Testing, Documentation, Code Review.